

Pseudocode as Scientific Planning

Week 2, Day 2

BIO/CHEM 806

Welcome!

Today: Pseudocode

Pseudocode = Writing down what you plan to do before you do it

Not intimidating! Just planning.

Last Class Recap

Monday: Organizing projects

Folders, file structures, relative paths, READMEs

Today: How do you plan what goes *inside* those scripts?

Pseudocode bridges scientific thinking and code

Pseudocode: Your Recipe

Artwork by Allison Horst

Planning Before Cooking

Artwork by Allison Horst

Just like a chef plans their recipe before cooking:

Pseudocode = Your recipe (the plan)

Code = Cooking the meal (execution)

R/RStudio = Your kitchen and tools

Data = Your ingredients

You wouldn't start cooking without a recipe!

What Is Pseudocode?

Step-by-step description of what you want your code to do, written in plain language

- Not code
- Not prose
- Something in between

Think: Recipe for analysis

Clear enough someone could follow it

Specific enough = no ambiguity

General enough = not worrying about syntax

Pseudocode Example

Bad (too vague): - "Clean the data"

Better (pseudocode):

- "Load temperature data from CSV file"
- "Remove rows where temperature is missing"
- "Remove rows where temperature < -50 or > 60"
- "Calculate mean temperature for each site"
- "Save cleaned data to `data/processed/temp_clean.csv`"

See the difference?

Why Pseudocode Matters

1. Forces you to think before you code

Most mistakes = typing before knowing what you're doing

2. Reveals ambiguity

"Clean the data" seems clear... until you try to write pseudocode

3. Makes collaboration easier

Review logic before spending hours coding

Key insight: Good pseudocode translates to any programming language

Pseudocode vs. Code vs. Prose

Prose (too informal): > "Remove bad data and make a graph"

Pseudocode (just right):

- Load temperature data
- Remove rows where temperature < -50 or > 60
- Create scatterplot of temperature vs. elevation
- Save plot to `figures/temp_elevation.png`

Code (too specific):

```
1 data <- read.csv("data/raw/temps.csv")
2 data_clean <- data %>% filter(temperature >= -50, ...)
```

What Makes Good Pseudocode

Good pseudocode has:

- ✓ **Verbs** — Load, Remove, Calculate, Create, Save
- ✓ **Specific conditions** — “temperature < -50”, “absorbance > 3”
- ✓ **Clear outputs** — “scatterplot”, “save to figures/”
- ✓ **Domain knowledge** — “out of linear range”, “GPS error”

Detailed enough = unambiguous

General enough = not worrying about syntax

Common Pseudocode Mistakes

Mistake 1: Too vague

✗ "Standardize the measurements"

✓ "Divide each mass by wing length to get body condition index"

Mistake 2: Too detailed (writing code)

✗ "Use filter() where count >= 5"

✓ "Remove sites with fewer than 5 observations"

Mistake 3: Skipping steps

✗ "Load data, make plot"

✓ "Load → Remove invalid → Calculate mean → Make bar plot"

Mistake 4: No decision points

✗ "Remove outliers"

✓ "Remove values more than 3 SD from mean"

Let's Practice Together

Scenario: Dataset of bird observations

Each row has:

- Species name
- Date
- Location (lat/long)
- Observer name

Goal:

1. Count observations per species
2. Bar chart of top 10 species
3. Save the chart

Let's build pseudocode step by step!

Building Pseudocode Together

Type in chat: What's the first step?

[Wait for responses]

STEP 1: Load the bird observation data

Next?

STEP 2: Count the number of observations for each species

Our Complete Pseudocode

STEP 1: Load the bird observation data
STEP 2: Count the number of observations for each species
STEP 3: Sort species by count (highest to lowest)
STEP 4: Keep only the top 10 species
STEP 5: Create bar chart (species on x-axis, count on y-axis)
STEP 6: Save chart to figures/top_species.png

- ✓ Clear verbs
- ✓ Specific steps
- ✓ Logical order
- ✓ Not code
- ✓ Explicit output

Spotting Ambiguity: Example 1

What's wrong with this?

```
Load the dataset  
Clean the data  
Analyze it  
Save the results
```

Type in chat: What's ambiguous?

[Wait for responses]

Problem: Too vague! What does “clean” mean? What analysis? What results?

Spotting Ambiguity: Example 2

What's wrong?

```
Load temperature data  
Remove outliers  
Calculate average temperature  
Make a plot
```

Type in chat: What's missing?

[Wait for responses]

Problem: How do we define outliers? Average by what? What kind of plot?

Better Version

```
Load temperature data from field sensors
Remove temperatures < -50°C or > 60°C (sensor errors)
Remove temperatures > 3 SD from site mean (outliers)
Calculate mean temperature for each site
Create bar plot of mean temperature by site
  (ordered north to south)
Save plot to figures/temp_by_site.png
```

Much clearer! Every decision is explicit.

Breakout Activity - 15 minutes

Task: Rewrite your Week 1 analysis as formal pseudocode

In Week 1, you described cleaning a dataset with missing/invalid values

Now:

1. Rewrite as numbered pseudocode steps
2. Be specific — no vague verbs
3. Make decisions explicit
4. Write in Google Doc (link in chat)

When done: Pseudocode someone else could follow exactly

The Scenario

You're analyzing bee diversity data from urban gardens. Your dataset has a "Garden Type" column with:

- Community garden
- Community Garden
- community garden
- Comm. garden
- School garden
- school garden (community-maintained)
- Private yard
- private yard (open to public)

Task: Write a decision log entry explaining how you would standardize these categories — and why.

Model Decision Log Entry

Decision log: Standardize `Garden Type` categories

Problem: Same garden type appears with different capitalization, abbreviations, and parenthetical notes → creates fake extra categories.

Decision: Create a new column `garden_type_std` with canonical values: - `community_garden`
- `school_garden` - `private_yard`

Rules: 1. Trim whitespace + convert to lowercase

2. Remove parenthetical notes (e.g., "(community-maintained)")

3. Treat "comm." as "community"

4. Map by keywords: - contains "community" → `community_garden` - contains "school" → `school_garden` - contains "private" or "yard" → `private_yard` 5. Otherwise → `unknown` + flag for review

Why: Keeps the *scientific comparison* (type of garden) consistent and reproducible; extra notes can become a *separate variable later*.

Pseudocode: Normalize Garden Type

FOR each row in dataset:

Garden Type (working copy)

→ trim whitespace

→ convert to lowercase

→ remove text in parentheses

→ replace "comm." with "community"

END

Pseudocode: Classify Garden Type

FOR each row in dataset:

IF Garden Type contains "community":

garden_type_std = "community_garden"

ELSE IF Garden Type contains "school":

garden_type_std = "school_garden"

ELSE IF Garden Type contains "private" OR "yard":

garden_type_std = "private_yard"

ELSE:

garden_type_std = "unknown"

flag row for review

END

QA Check: Did We Do It Right?

Never standardize silently. Always check your work.

1. Create a table showing:

- original `Garden Type.` | new `garden_type_std`

2. Verify:

- Only the expected categories appear
- Similar values map to the same category
- `unknown` values are rare and explainable

3. If rules change:

- Update the decision log

Pseudocode as Debugging Tool

Pseudocode isn't just for planning—it's for debugging!

When code fails, two possibilities:

1. **Logic is flawed** — asking computer to do wrong thing
2. **Syntax is wrong** — asking correctly but wrong words

With pseudocode: - Check: "Is this what I actually want to do?" - If pseudocode is right → syntax problem (easier!) - If pseudocode is wrong → fix logic first, then code

Saves hours of frustration

Before Monday

Exit Ticket: Write pseudocode (5-8 steps) for ONE of these:

Option A: Plant heights at different light levels - Remove negative heights (errors) - Remove heights > 300 cm (errors) - Calculate mean height per light level - Save summary to file

Option B: UV-Vis kinetics data - Subtract baseline (first 5 timepoints) - Remove absorbance < 0 or > 3 (out of range) - Calculate slope (reaction rate) - Save rate constant

Post in discussion board by Monday!

Exit Ticket (Canvas) - 2-3 minutes

Complete before leaving:

1. **Define:** What is pseudocode? How is it different from code? (2-3 sentences)
2. **Critique:** What's wrong with this pseudocode?

```
Process the samples  
Do statistics  
Make graph
```

3. **Write:** Pseudocode for calculating BMI from height and weight data (4-5 steps, be specific about units and formulas)

Submit in Canvas →

Key Takeaways

1. **Pseudocode = planning tool** bridging thinking and code
2. **Good pseudocode = specific, ordered, explicit**
3. **Pseudocode reveals flawed logic** before coding
4. **Language-agnostic** — translates to any programming language
5. **Use domain knowledge** — measurement ranges, constraints

Next week: Start writing actual code in Quarto!

Your pseudocode this week makes that easier.

See you Monday!